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Performance control in handstands: challenging entrenched coaching strategies for young gymnasts

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ABSTRACT

Despite well-accepted knowledge concerning the importance of the wrists' contributions in handstand balances, common handstand coaching strategies prioritise developing the required aligned and aesthetic body shape. Thus, this study investigated the effects of a sole postural training on the maintenance compared to the execution quality in handstands in young competitive gymnasts. Therefore, twenty-two participants (13 females, aged: 8.54 ± 1.56 years; 9 males, aged: 8.67 ± 0.87 years) were randomly assigned into two counterbalanced groups, GPT (guided postural training) and CON (control), each $n = 11$. Within one week, GPT received four standardised training sessions (T1, T2, T3 and T4) including verbal and tactile advices addressing a straight body posture. Prior to T1 (pre-test), after T2 (mid-test) and after T4 (post-test), all participants (GPT and CON) were asked to perform five handstands where handstand executions were assessed by four gymnastics judges based on standardised video recordings which also captured the balance time. There were no group-related changes in balance time ($p > .05$); however, following T2, results revealed a positive impact of GPT on postural execution ($p < .05$). With respect to the importance of forearm activity in handstand performances, further research is needed approaching wrist-related advice in handstand acquisition.

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Gymnastics; inverted stance; balance; postural execution; feedback; children

Introduction

In gymnastics, developing the handstand is a well-accepted fundamental training content that is considered to be of utter importance right from the beginning of a young gymnasts' career (Uzunov, 2008). Across all apparatus in female and male artistic gymnastics, the handstand serves as a starting position or as a static finish of movement techniques using the hands as body-apparatus interface (Hedbávný, Bago, & Kalichová, 2013; Hedbávný, Sklenaříková, Hupka, & Kalichová, 2013). The varying mechanical properties of the different apparatus complicate the postural correcting and require multiple experiences and specific muscle activation (Kochanowicz et al., 2017).

Motor learning progresses in handstand acquisition can be assessed according to different criteria. Firstly, competitive gymnasts aim to acquire an aesthetic and steady body shape characterised by the axial alignment of the hands, shoulders, hip and feet (Gerling, 2009; Uzunov, 2008). Apart from that, obviously, handstand performances can be determined by examining the maintenance of balance during inverted stance. Both aspects have received considerable attention in current biomechanical and motor behavioural research addressing a transfer of postural control strategies used in the upright stance to handstand balancing movement patterns (Blenkinsop, Pain, & Hiley, 2017; Gautier, Marin, Leroy, & Thouwarecq, 2009; Gautier, Thouwarecq, & Chollet, 2007; Kerwin & Trewartha, 2001; Kochanowicz et al., 2017; Mohammadi & Yazici, 2016; Sobera, Siedlecka, Piestrak, Sojka-Krawiec, & Graczykowska, 2007; Yeadon & Trewartha, 2003).

Previous studies using a force plate or capturing neuromuscular activation via surface electromyography (EMG) provide evidence that emerging anterior-posterior sway during the upright stance has been corrected by the ankle extensors (Gatev, Thomas, Kepple, & Hallett, 1999). Using this pattern of muscular activation, the human body resembles a single-segment inverted pendulum rotating about the ankle joints providing that the centre of gravity moves within the centre of pressure excursion limits (Kuo & Zajac, 1993; Runge, Shupert, Horak, & Zajac, 1999; Stephens, Frank, Burleigh, & Winter, 1992). In response to insufficient ankle torques caused by extensive or fast perturbations, the “ankle strategy” is supported by the “hip strategy” that is characterised by the activation of the hip extensors and flexors, respectively (Runge et al., 1999). In light of these findings, it seems reasonable that selected postural control strategies prefer the employment of the most inferior joint before, if necessary, involving more superior joints torques.

It is well-accepted that the single segment inverted pendulum control strategy in erect stance is also applicable to describe the balance mechanisms used in maintaining the handstand on a plane surface (e.g. Blenkinsop et al., 2017). Despite apparent similarities compared to normal stance, the mechanical characteristics of the handstand are determined by a reduced support area and an increased distance of the centre of gravity to this support area due to an additionally involved joint (Hedbávný et al., 2013a, 2013b). However, findings about postural activities during handstand maintenance are in line with previous research concerning balance strategies in the erect position (Gautier et al., 2007; Kerwin & Trewartha, 2001; Sobera et al., 2007). Accordingly, resuming the findings of Runge et al. (1999) and Kerwin and Trewartha (2001) revealing predominant torques at the interface between body and support surface, wrist-controlled corrections of the centre of gravity are suggested to be of exceedingly importance for well-balanced handstand performances. Moreover, there is evidence that inadequate handstand balances involve the “hip strategy” to compensate disturbances (Gautier et al., 2009).

In view of these biomechanical reports, well-accepted motor learning concepts according to the handstand acquisition have to be elucidated. In general, there is widespread and non-sport-specific acceptance that motor learning is advanced due to external information prior to and after practice, considering, for example, movement visualisation as well as verbal instructions and tactile advice (Schmidt & Lee, 2005). These methods provide augmented feedback about the movement in addition to the inherent information received by sensory mechanisms (Schmidt & Lee, 2005). Moreover, despite generally positive influences of augmented feedback on movement learning, declarative knowledge may also have disruptive

effects on relatively skilled performances due to impaired automated behaviours (Masters & Maxwell, 2008). With regard to the children's cognitive efforts while receiving feedback, age-dependent differences in selective attention and information processing should be taken into consideration when assessing the amount of motor learning progresses (Chuah & Maybery, 1999; Ferguson & Bowey, 2005; Tipper, Bourque, Anderson, & Brehaut, 1989).

Sport-specific approaches regarding motor learning suggest combined movement practicing and movement observing to increase motor learning progresses (Hayes, Hodges, Scott, Horn, & Williams, 2006). With respect to sport-specific research on motor learning in gymnastics, a few studies examined different strategies of augmented information to accelerate handstands' learning processes (Croix, Lejeune, Anderson, & Thouvarecq, 2010; Ghavami, Hosseini, Mohammadzadeh, Maleki, & Borhani, 2012; Maleki, Nia, Zarghami, & Neisi, 2010; Masser, 1993; Rohleder & Vogt, 2018). Following different approaches using, for example, video data (Masser, 1993; Rohleder & Vogt, 2018) or the judging of gymnastics experts (Maleki et al., 2010), these studies investigated the effects of tactile advice (Croix et al., 2010), verbal instructions (Masser, 1993), visual information (Ghavami et al., 2012) or even combined feedback strategies (Maleki et al., 2010; Rohleder & Vogt, 2018) on the handstand performance. For example, Croix et al. (2010) showed that light finger touch applied to the thigh facilitates the maintenance in handstands. Moreover, Rohleder and Vogt (2018) observed improved postural executions in handstands after receiving commented tactile advice compared to a visual-comparative feedback.

In addition, current gymnastics practical training literature provides comprehensive methodology addressing a high-quality gradual learning of the handstand (Bessi, 2009; Gerling, 2009; Uzunov, 2008). Except for Uzunov (2008), nevertheless, it is apparent that well-established applied coaching methods for teaching the handstand to young gymnasts almost exclusively relate to an improved aligned body shape, aiming at a high-quality execution and postural performance (Bessi, 2009; Gerling, 2009). Indeed, besides numerous methodological steps with regard to the handstand posture, Bessi (2009) has presented only one aspect broaching the issue of wrist and finger work. Although aware of the importance of a steady body shape, the study of, for example, Kerwin and Trewartha (2001) raises doubts about the benefits on the exclusive teaching of an aligned posture in order to improve balance abilities in handstand performances. In view of the wrists' importance for handstand balances, one can ask the questions why so little attention is given to that in practical coaching and whether exclusive postural practices without explicit cues regarding the wrist-work are sufficient in improving the maintenance vs. execution quality in handstands.

Therefore, the purpose of this study is to challenge the importance of guided training on postural issues in improving the performance control of a handstand, taking the maintenance compared to the execution quality in young competitive gymnasts into account. Following an applied approach, it is hypothesised that sole postural exercises accompanied by augmented information (1) do increase the time of maintenance (Croix et al., 2010) and (2) improve the postural execution quality in handstands (Rohleder & Vogt, 2018). With respect to Gautier et al. (2009), it is further hypothesised that (3) the balance time correlates with the execution quality in handstand performances.

Methods

Participants

Twenty-two young healthy gymnasts (13 females, age: 8.54 ± 1.56 years; height: 129.40 ± 9.02 cm; weight: 26.74 ± 4.86 kg; 9 males, age: 8.67 ± 0.87 years; height: 130.78 ± 6.63 cm; weight: 27.40 ± 5.16 kg) volunteered to participate in this study, after parents gave written informed consent. The University's Ethics Committee approved this study. The sample size was characterised by only including competitive gymnasts between seven and eleven years of age. In terms of specifying the participants' level, the youngest gymnasts (aged: 7–8 years) take part in regional competitions and may be described as novices. Moreover, advanced gymnasts (aged: 9–11 years) even have to meet national requirements. At the age of nine, the handstand on the floor is for the first time part of the required athletic standards which are regularly monitored in female and male squad athletes by the national gymnastics federation (Deutscher Turner-Bund, 2015a, 2015b). Participants who had trained gymnastics for less than half a year were excluded from this study. This at least guaranteed a necessary minimum level of movement experience in performing the lunge entry and upward swing to handstand in terms of preventing overstrained demands.

Experimental protocol

The participants were randomly assigned into one of two counterbalanced groups: (1) guided postural training on aligned steady posture (GPT, $n = 11$); (2) control group without any additional training (CON, $n = 11$). Within one week of regular gymnastics training (i.e. novice gymnasts: 4 training sessions for about 3 h; advanced gymnasts: 6 training sessions for about 3 h), the GPT group performed an additional standardised fifteen-minute training programme in order to develop a high-quality (= aesthetic and strong) posture throughout the handstand. While GPT received this training programme four times (T1, T2, T3 and T4) during the week, CON continued normal training as usual. Based on the same content at the beginning of each GPT session, the level of difficulty slowly increased from T1 till T4. Inspired by Uzunov (Uzunov, 2008), the instructions and exercises referred to the following three essential properties of postural excellence, (1) flattening of the lower back by posterior pelvis rotation (verbal instruction: “rotate your pelvis!”; tactile advice: light fingertip contact on the gluteus maximus muscle). (2) Protraction of the scapula (verbal instruction: “chest in!”; tactile advice: light fingertip on the sternum). (3) Fixing the shoulders open by scapular elevation (verbal instruction: “squeeze your ears!”; tactile advice: light arm pulling). Working on these points is reported to provide a stabilising effect whilst in the handstand position (Uzunov, 2008). Based on this concept, the standardised training session was designed using static shape holds in prone and press-up positions as well as aligned holds with the back pressed against the wall (Uzunov, 2008). These static exercises were accompanied by dynamic adjustments regarding pelvis rotation and scapula protraction and elevation. In line with given volume recommendations, the static positions were performed for 30 s within 3 sets including 5 repetitions of shoulder and pelvis work (Uzunov, 2008). All exercises were complemented by verbal instructions and tactile advice as well as practical exercise demonstrations.

Prior to T1 (pre-test = M1), after T2 (mid-test = M2) and after T4 (post-test = M3), each participant (GPT and CON) performed five trials of the lunge entry and upward swing to

handstand on the gymnastics floor, aiming at fixing this inverted stance position for as long as possible. Comparable preload conditions were ensured by scheduling the tests at the beginning, but after the collective warm up of the training sessions. Not prior to the pre-test (M1), but before M2 and M3, only GPT participants watched a commented video showing a perfectly performed handstand to receive another visual information with the objective of additive observational learning effects (Ghavami et al., 2012; Maleki et al., 2010). Prior to the tests, all GPT and CON participants performed a single previous attempt of swing up to handstand to serve as familiarisation. During the tests, the participants were requested to follow the instructions received during the training sessions as well as to consider their observations in the video sequence at their possible best. In order to examine the balance coordination abilities instead of strength endurance, the maintenance of fifteen seconds was defined as the point when the test could be stopped. Moreover, participants got thirty seconds between the test trials to recover.

Data collection

Following the practical approach that may be copied by coaches or teachers, the handstands were recorded in the sagittal plane for each trial using a tablet PC (Apple iPad Air, frequency 60 Hz, resolution 1080p). The tablet was fixed using a tripod at a height of one metre and at a distance of four metres to the middle of the support surface. The positions of the tripod and the support surface were indicated using marker points ensuring that the captured scene was identical in each training session. For the primary outcome measure, the balance time was monitored to compare GPT and CON. Using the Kinovea 0.8.15 software, the criteria for the measuring time period during handstand was defined by a corridor of 15° to each side of the vertical line above the wrists. Time measuring was prematurely terminated when either the corridor was left by the feet, a rolling over movement set in or the placing back of the feet was initiated. Corrective spreading or straddling of the legs (counting the moment of the feet gap) as well as excessive bending of the elbows were forbidden, leading to a termination of the time measuring.

Inspired by Maleki et al. (2010), postural execution of each handstand trial was assessed by four independent artistic gymnastics experts, who were certified as international judges in the Olympic cycle of Tokyo 2020 by the International Gymnastics Federation (FIG). Watching all video clips in randomised order, the judges assigned scores to each trial using a Likert scale between 0 and 10 points considering the official performance criteria of the current FIG's code of points (Fédération Internationale de Gymnastique, 2017). Mean scores were calculated to quantify the execution quality of the handstands performed in M1, M2 and M3.

Statistical analysis

For balance time and postural execution, repeated measures analyses of variance (ANOVA) were computed to display main and interaction effects within factors group (GPT vs. CON) and time of measurement (M1, M2 and M3). Moreover, repeated measures analyses of covariance (ANCOVA) were calculated entering age as a covariate to obtain possible effects of age. Furthermore, repeated measures ANOVA were computed separately for novice ($n = 9$; aged: 7–8 years) and advanced gymnasts ($n = 11$; aged: 9–11 years) to determine age-related

influences of GPT and CON in more detail. Fisher's least significant difference (LSD) served as *post hoc* test. In order to ensure interrater reliability of the expert judgements, the intraclass correlation coefficient (ICC) was calculated using the two-way mixed effects model for measures of absolute agreement. Furthermore, possible correlations between balance time and postural execution were determined using Pearson's correlation coefficient.

Data in the figures and tables are presented as mean (M) $\pm$ standard deviation (SD). The level of significance was set at $p < 0.05$. Data were excluded from analysis when the participant missed at least one test session or one training session (GPT) or when the perfection of the handstand was already apparent throughout M1, M2 and M3. In line with these criteria, a sample size of $n = 20$ remained for statistical analyses ($n = 10$ for each group).

Results

Balance time

Repeated measures ANOVA revealed no interactions between time of measurement and group, $F(2, 36) = 0.71$, $p = .50$, $\eta^2 = .04$. Further, no main effects were obtained for time of measurement $F(2, 36) = 2.05$, $p = .14$, $\eta^2 = .10$, and for group, $F(1, 18) = 0.08$, $p = .78$, $\eta^2 = .00$ (Figure 1(A)).

Repeated measures ANCOVA revealed no interactions between time of measurement and group, $F(2, 34) = 1.07$, $p = .35$, $\eta^2 = .06$, whereas significant interactions were obtained between time of measurement and age, $F(2, 34) = 5.68$, $p = .01$, $\eta^2 = .25$. Further, main effects were obtained for time of measurement $F(2, 34) = 4.53$, $p = .02$, $\eta^2 = .21$, but not for group, $F(1, 17) = 0.05$, $p = .82$, $\eta^2 = .00$.

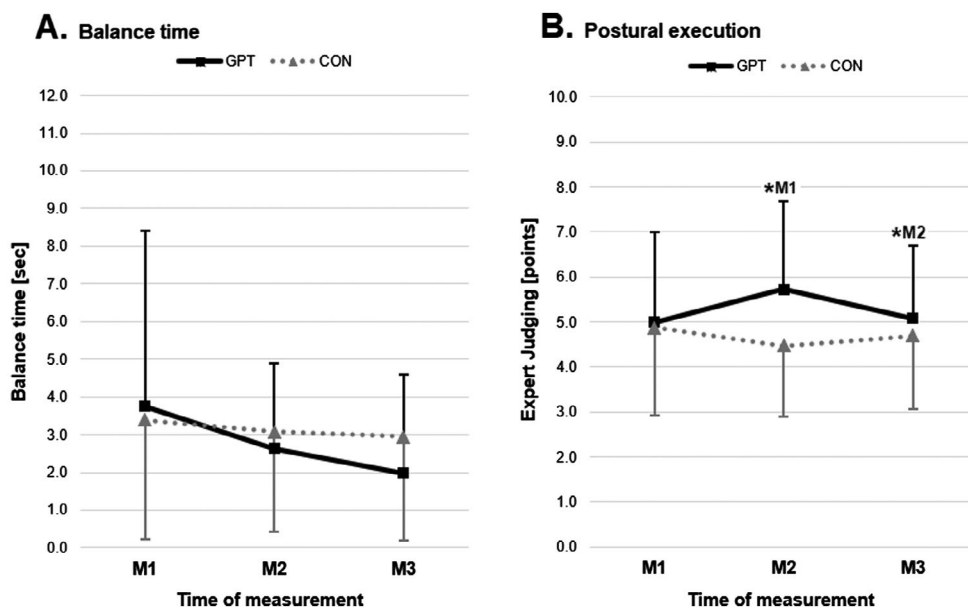


Figure 1. Interaction plot (repeated measures ANOVA) showing the (A) handstands' mean balance times and (B) mean scores for postural execution during the pre- (M1), mid- (M2) and post-test (M3) in GPT and CON. Displayed are means with SD error bars. Level of significance including reference: ** $p < .01$; * $p < .05$.

Post hoc analysis revealed significant changes from M1 to M3 ($p = .05$). In GPT, significant decreases are revealed from M1 to M3 ($p = .03$; Table 1), whereas no significant changes were obtained in CON (all $p > .05$; Table 1).

Separate repeated measures ANOVA revealed no interactions between time of measurement and group for novice gymnasts, $F(2, 14) = 0.07$, $p = .93$, $\eta^2 = .01$, and for advanced gymnasts, $F(2, 18) = 1.18$, $p = .33$, $\eta^2 = .12$. Further, no main effects were obtained for time of measurement in novice gymnasts, $F(2, 14) = 0.19$, $p = .83$, $\eta^2 = .03$, and in advanced gymnasts, $F(2, 18) = 2.81$, $p = .09$, $\eta^2 = .24$ (Figures 2(B) and (C); Table 1).

Postural execution

A high degree of reliability was found between the expert judgements. The average measure ICC was .946 with a 95% confidence interval from .928 to .959, $F(299, 897) = 21.36$, $p < .001$.

Repeated measures ANOVA revealed significant interactions between time of measurement and group, $F(2, 36) = 4.47$, $p = .02$, $\eta^2 = .20$ (Figure 1(B)). Further, no main effects were obtained for time of measurement $F(2, 36) = 0.63$, $p = .54$, $\eta^2 = .03$, and for group, $F(1, 18) = 0.53$, $p = .48$, $\eta^2 = .03$.

Repeated measures ANCOVA revealed significant interactions between time of measurement and group, $F(2, 34) = 4.25$, $p = .02$, $\eta^2 = .20$, whereas no significant interactions were obtained between time of measurement and age, $F(2, 34) = 0.72$, $p = .49$, $\eta^2 = .04$. Further, no main effects were obtained for time of measurement, $F(2, 34) = 0.89$, $p = .42$, $\eta^2 = .05$, and for group, $F(1, 17) = 1.18$, $p = .29$, $\eta^2 = .07$.

Post hoc analysis revealed significant increases in GPT from M1 to M2 ($p = .03$) and significant decreases from M2 to M3 ($p = .04$; Table 1), whereas no significant changes were obtained in CON (all $p > .05$; Figure 1(B); Table 1).

Separate repeated measures ANOVA revealed significant interactions between time of measurement and group for advanced gymnasts, $F(2, 18) = 11.07$, $p = .00$, $\eta^2 = .55$, but not for novice gymnasts, $F(2, 14) = 0.52$, $p = .60$, $\eta^2 = .07$. Further, main effects were obtained for time of measurement in novice gymnasts, $F(2, 14) = 3.93$, $p = .04$, $\eta^2 = .36$, but not in advanced gymnasts $F(2, 18) = 0.04$, $p = .95$, $\eta^2 = .01$ (Figures 2(C) and (D); Table 1).

For advanced gymnasts, *post hoc* analysis revealed significant increases in GPT from M1 to M2 ($p = .00$), whereas significant decreases were obtained in CON ($p = .00$). Further, *post hoc* analysis revealed significant increases in CON for novice gymnasts from M1 to M2 ($p = .03$) and significant decreases from M2 to M3 ($p = .01$; Figures 2(C) and (D); Table 1).

Correlations

The correlation analysis revealed significant positive correlations between balance time and postural execution at all three times of measurement for GPT (M1: $r(8) = .82$, $p = .00$; M2: $r(8) = .90$, $p = .00$; M3: $r(8) = .85$, $p = .00$) and CON (M1: $r(8) = .76$, $p = .01$; M2: $r(8) = .77$; $p = .01$; M3: $r(8) = .80$; $p = .01$; Figure 3).

Discussion

The present study investigated the effects of an exclusive guided postural training (GPT) on the maintenance compared to the execution quality of handstands in young competitive

Table 1. Changes in balance time and postural execution for GPT and CON conditions (in total as well as separated by age groups) during the pre- (M1), mid- (M2) and post-test (M3).

Group	<i>n</i>	Balance time [sec]			Expert judging [points]		
		M1	M2	M3	M1	M2	M3
		M ± SD	M ± SD	M ± SD	M ± SD	M ± SD	M ± SD
GPT	total ^a	3.75 ± 4.65	2.64 ± 2.24	1.98 ± 2.61 ^{*M1}	5.00 ± 2.01	5.72 ± 1.97 ^{*M1}	5.08 ± 2.04 ^{*M2}
	novice gymnasts ^b	1.31 ± 0.83	1.42 ± 1.01	1.39 ± 1.72	4.22 ± 1.76	4.49 ± 1.30	4.15 ± 1.51
	advanced gymnasts ^b	6.19 ± 5.76	3.87 ± 2.56	2.57 ± 3.39	5.79 ± 2.12	6.94 ± 1.82 ^{**M1}	6.01 ± 2.22
CON	total ^a	3.39 ± 3.17	3.08 ± 2.67	2.94 ± 2.74	4.87 ± 1.95	4.47 ± 1.56	4.70 ± 1.63
	novice gymnasts ^b	0.73 ± 0.59	0.96 ± 0.63	0.70 ± 0.30	3.22 ± 1.05	3.88 ± 0.94 ^{*M1}	3.27 ± 0.55 ^{*M2}
	advanced gymnasts ^b	5.17 ± 2.91	4.48 ± 2.58	4.43 ± 2.60	5.97 ± 1.60	4.87 ± 1.84 ^{**M1}	5.64 ± 1.38

^aDisplayed are LSD-adjusted pairwise comparisons computed as *post hoc* after conducting repeated measures ANCOVA.
^bDisplayed are LSD-adjusted pairwise comparisons computed as *post hoc* after conducting separate repeated measures ANOVA for novice and advanced gymnasts. Level of significance including reference.
****p* < .01; **p* < .05.

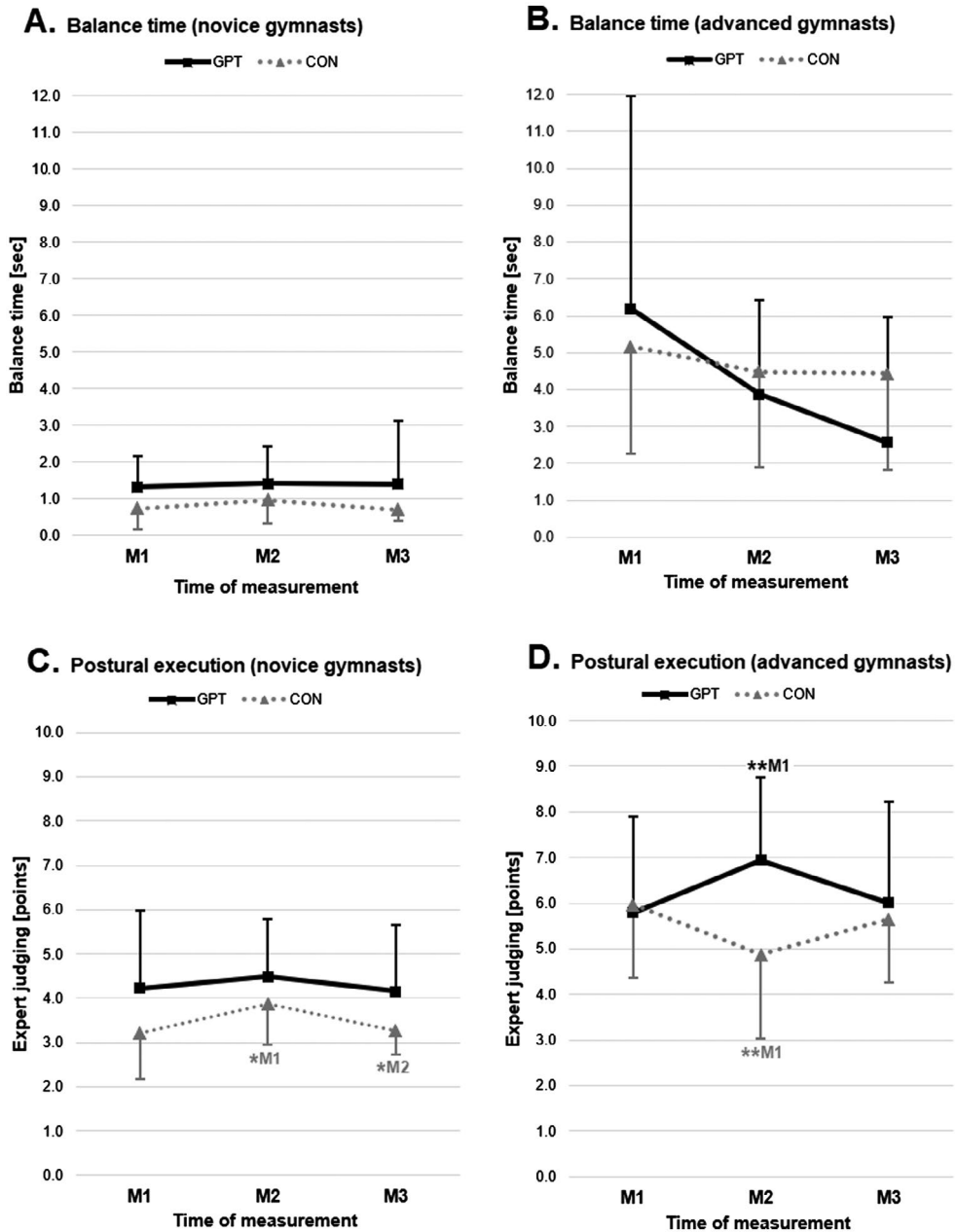


Figure 2. Displayed are the handstands' mean balance times in GPT and CON for (A) novice gymnasts (aged: 7–8 years) and (B) advanced gymnasts (aged: 9–11 years) during the pre- (M1), mid- (M2) and post-test (M3); displayed are the mean postural execution scores in GPT and CON for (C) novice gymnasts and (D) advanced gymnasts during the pre- (M1), mid- (M2) and post-test (M3). Level of significance including reference: ** $p < .01$; * $p < .05$.

gymnasts. At no time, GPT induced differences in balance time compared to controlled conditions (CON), whereas the groups showed different changes in postural execution. Divided by age groups, advanced gymnasts' postural execution increased after two sessions

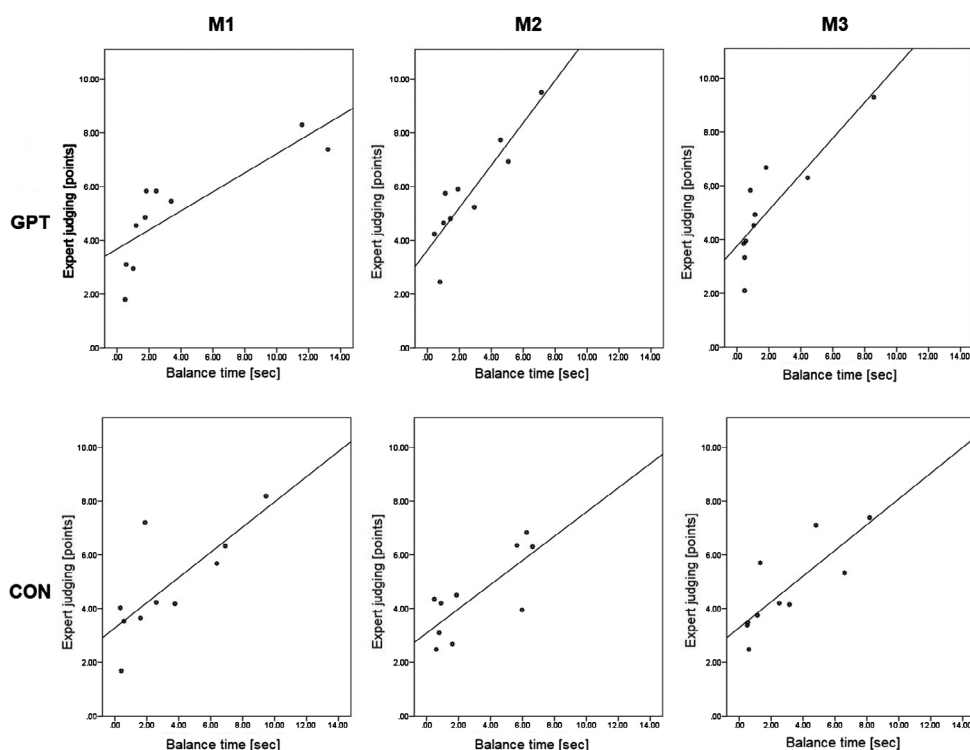


Figure 3. Positive correlations between the balance time [sec] and the postural execution [points] during the pre- (M1), mid- (M2) and post-test (M3) in GPT and CON.

of GPT, but decreased in CON during the same period. Novice gymnasts' balance time and postural execution remained unaffected by GPT.

There was no positive effect of GPT on young gymnasts' maintenance in handstands, contradicting our initial hypothesis (1) that postural exercises complimented by augmented information extend young gymnasts' time of maintenance in handstands. Thus, from the balance abilities' perspective, the present findings are in contrast to previous reports suggesting that movement practising supported by augmented feedback information (e.g. tactile advice or visual information) improves sport-specific skill acquisition (Schmidt & Lee, 2005) and, in particular, handstand acquisition (Croix et al., 2010; Ghavami et al., 2012; Hayes et al., 2006; Maleki et al., 2010; Masser, 1993; Rohleder & Vogt, 2018). We attribute these findings to two according interpretative approaches.

Firstly, from a biomechanical point of view, numerous studies reveal that balance mechanisms during inverted stance on the floor cannot be exclusively attributed to movement patterns providing an aligned steady body posture, but also have to take the contributions by the wrists into account (e.g. Blenkinsop et al., 2017; Gautier et al., 2007, 2009; Hedbávný et al., 2013, 2013; Kerwin & Trewartha, 2001; Mohammadi & Yazici, 2016; Sobera et al., 2007). Although generally improved handstand performances require strengthened core muscle activation (e.g. Bessi, 2009; Gerling, 2009; Kochanowicz et al., 2017; Uzunov, 2008), there is evidence that well-balanced handstands, particularly if performed by high-level gymnasts, have been primarily controlled with the wrist joints (Gautier et al., 2009; Kerwin

& Trewartha, 2001). Larger perturbations and displacements of the centre of gravity require assisting corrections in the shoulder joints (Gautier et al., 2007; Kerwin & Trewartha, 2001). The same reports have detected that only less successful handstand performances are influenced by increasing hip torques (Gautier et al., 2009; Kerwin & Trewartha, 2001). With respect to this well-accepted knowledge, the present study's findings support initial doubts whether exclusive coaching of an aesthetic and strong posture is sufficient to improve necessary balance abilities in handstands.

Secondly, the present findings seem to be in line with the reinvestment theory by Masters and Maxwell (2008) suggesting that augmented information and movement-related knowledge may disrupt relatively skilled performances. Although failing significance, the even impaired balance times following GPT that were particularly observed in advanced compared to novice gymnasts may be attributed to older gymnasts' advantageous capabilities for information processing and attentional control (Chuah & Maybery, 1999; Ferguson & Bowey, 2005; Tipper et al., 1989). Enhanced cognitive capacities may enable older gymnasts to try out short-term changes in the postural adjustment of joint angles during handstand, presumably at the expense of an unknown postural perception (Rohleder & Vogt, 2018) and, as a consequence, initially decreased balance abilities. Thus, it seems reasonable to assume that the additional declarative knowledge made the advanced gymnasts become conscious of the relevant corrective movements that they have to do when performing an aligned handstand. Though surprising at first, these circumstances may cause impaired handstand balances in the more experienced gymnasts who may already have developed automated motor processes to control their handstand performances (Masters & Maxwell, 2008). With this and with respect to the relatively short period for intervention, the present study's findings may raise the awareness for possible wrist-related advice as well as for the timing of qualitative training measures considering the possible negative impact on other performance factors.

Furthermore, the present study revealed divergent changes for postural execution in GPT and CON. The enhanced postural performances after two sessions of GPT compared to impaired postural performances in CON are in line with our second hypothesis (2). Interestingly, subsequent to the final training session, there were no differences between the groups which then contradict our hypothesis (2). However, enhanced postural performances in the mid-test are consistent with previous reports revealing that augmented information such as tactile advice, visual information and precise verbal cues have facilitated children's, novices' and experts' handstand skill acquisition, respectively, (Croix et al., 2010; Ghavami et al., 2012; Maleki et al., 2010; Masser, 1993) even, at least in parts, within short-term approaches (Rohleder & Vogt, 2018). In view of this extensive knowledge supporting the positive effects of different feedback strategies on handstand performances, it can be reasonably assumed that the applied multi-factorial intervention design used in the present study has positively influenced the execution quality of the performed handstands in the mid-test. This approach of interpreting, at least, would be in line with the findings of Hayes et al. (2006) displaying enhanced motor learning due to combined movement practising and observing. With respect to this and to the previously discussed results for balance time, this interpretation is attributed to the continuative hypothesis that different feedback strategies influence several issues of motor behaviour in handstands, in this case balance time and postural execution, in a different manner. Thus, the applied feedback strategy has been suggested to positively affect the postural adjustment of the large body joints, but has

not influenced motor behaviour in terms of enhanced balance abilities. However, absent execution differences between groups after one week of training have to be discussed, likely in relation to the rising physical strain, which would explain the observed failure of the GPT participants, implementing the received instructions to enhance their handstand performances during the post-test. Furthermore, taking the observed age-related differences into account, enhanced postural performances in GPT relate to advanced gymnasts exclusively. The postural execution of advanced gymnasts increased in the mid-test, what might confirm the interpretation of the previous discussed findings concerning balance time. With respect to this, GPT is suggested to enable experienced gymnasts' short-term enhancements of the postural alignment of joint angles during handstand accompanied by decreased hand balance control. In Accordance to balance time, postural execution decreased in GPT up to the post-test, presumably due to additional general training load at the end of a week, what makes it, thus, even more difficult for children to implement received instructions. Finally, it can be assumed that the four sessions of additional training are not sufficient to sustainably enhance the postural execution quality, and less the maintenance in handstand, in young gymnasts. Considering that previous studies used several week-long approaches (Ghavami et al., 2012; Maleki et al., 2010; Masser, 1993), future studies on gymnastic-specific motor behavioural issues should think of extended intervention periods. However, on the other hand and with respect to the rapid development of demands in international gymnastics (Fédération Internationale de Gymnastique, 2017), the necessity of shortened skill learning times requires further research that aims to elucidate appropriate coaching strategies on imparting gymnastic skills most efficiently within a short period of time.

Further, in both groups, positive correlations between the balance time and postural execution were detected, which is in line with the third hypothesis. (3). With respect to well-known technical variants for maintaining the handstand position, one might also expect to observe varying balance durations regardless of the quality of the aligned handstand posture. Taking the biomechanical conditions into consideration, one may hypothesise that, for example, closed shoulder angles can be observed in novices' handstand performances because all deviations from the postural alignment result in a decreased height of the centre of gravity and, thus, facilitated equilibrium. However, taking the present results into account, it seems reasonable that these observations as well as the previously discussed divergent findings regarding balance time compared to postural execution may not reflect the fundamental interdependence between balance abilities and an aligned steady body shape in handstands. At least surprising by its magnitude, these correlations do not contradict previous reports revealing that less experienced gymnasts prefer patterns of hip muscular activity with high angular amplitudes to compensate centre of gravity disturbances (Gautier et al., 2009). Rather than using the movement pattern of a single-segment inverted pendulum as it was suggested to be appropriate in upright as well as in inverted stance (Blenkinsop et al., 2017; Gatev et al., 1999; Kuo & Zajac, 1993; Runge et al., 1999; Stephens et al., 1992), disturbed handstand balances are characterised by using the double-segment inverted pendulum control strategy (Gautier et al., 2009; Runge et al., 1999). This might be finally expressed by decreased judges' points for postural execution quality. Accordingly, well-balanced handstands are defined by predominant wrist torques to control the inverted stance posture without adjustments in further joints (Kerwin & Trewartha, 2001; Yeadon & Trewartha, 2003). Within this approach, future studies have to challenge the question on how to teach a "wrist-strategy" to novice gymnasts or young athlete gymnasts.

Limitations

Retrospectively and although further limiting the sample size of participants between seven and eleven years of age, we are well aware that the level of experience in balancing the handstand on the floor varied. However, this may be considered as an opportunity to possibly help specifying any age-related ranges in future research approaches. Further, it may be questioned whether purely maintenance information on handstands is sufficient when assessing an impact on balancing behaviour. However, with respect to the exceedingly applied objective, the chosen methods turned out to be useful, revealing crucial sport-specific implications that may be considered as transparent and, thus, may provide greater usability for scientifically interested gymnastic coaches.

Conclusion

The present study examined the impact of an exclusive postural training intervention on the maintenance and the quality of execution in young gymnasts' handstands. Postural training revealed slightly impaired maintenance in handstands which is more related to the challenge of implementing postural skills than just to the balance abilities. Common coaching strategies are suggested to consider the mutual dependences of balance and postural alignment to optimise comprehensive performance control in handstands. Regarding this, further research is necessary, particularly approaching the effects of wrist-related exercises and advice on the muscle activation in the respective limbs to further understand motor learning processes during handstand skill acquisition.

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